
CUMULATIVE LOGIT ORDINAL REGRESSION FOR SELF-REPORTED HEALTH IN NHANES

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Abstract

Self-reported health is an important measure in public health research because it shows how people describe their own general health. It is usually recorded in ordered categories such as Excellent, Very good, Good, Fair and Poor. Since these categories have a natural order, the response should be analysed using a method that respects this ordering.

This project uses cumulative logit ordinal regression to analyse self-reported general health using data from the National Health and Nutrition Examination Survey (NHANES). The analysis uses NHANES survey cycles 2009–2012 and studies the association of age, body mass index (BMI), gender and current smoking status with self-reported health. The final modelling dataset contains complete cases for all selected variables.

Both parallel and non-parallel cumulative logit models are fitted. The parallel model assumes that the effect of each predictor is the same across all cumulative thresholds. The non-parallel model allows predictor effects to vary across thresholds. The parallel model produced stable and interpretable results, including residual deviance, log-likelihood, AIC and BIC. The non-parallel model produced a slightly lower residual deviance, but the reduction was only 1.068 compared with the parallel model. This was very small compared with a deviance value above 7400.

The results suggest that cumulative logit ordinal regression is suitable for analysing ordered self-reported health outcomes. The parallel cumulative logit model was selected as the main model because it was simpler, more stable and easier to interpret. The non-parallel model was retained as a sensitivity check to show that relaxing the proportional odds assumption did not produce a meaningful improvement in model fit.

Introduction

Self-reported health is a common measure used in public health and health survey research. It is normally collected by asking people to describe their general health using categories such as Excellent, Very good, Good, Fair and Poor. Although the measure is subjective, it is useful because it reflects how people judge their own health condition. This judgement may be influenced by physical health, lifestyle, long-term illness, mental wellbeing and social factors.

The main aim of this project is to analyse self-reported general health using data from the National Health and Nutrition Examination Survey (NHANES). The analysis focuses on selected demographic and health-related predictors: age, body mass index (BMI), gender and current smoking status. The response variable is self-reported health, and the aim is to understand how the selected predictors are associated with the ordered health categories.

The research question for this project is:

How are age, BMI, gender and current smoking status associated with ordered self-reported general health in the NHANES data?

The response variable is ordered, so ordinal regression is suitable for this research question. Excellent is the highest health category and Poor is the lowest health category. For example, one cannot assume that the difference between Excellent and Very good is the same as the difference between Fair and Poor. Therefore, the outcome should not be treated as a continuous numerical variable. This follows the general idea of ordered regression models, where the response has ranked categories and the modelling approach should preserve that order (McCullagh 1980; Agresti 2010; Fullerton and Xu 2016; Course Material 2026a).

This project uses cumulative logit ordinal regression. Two versions of the model are fitted: the parallel model and the non-parallel model. The parallel model assumes that the predictor effects are the same across cumulative thresholds. The non-parallel model allows the predictor effects to differ across thresholds. The two models are compared using model output, residual deviance, fit statistics where available, predicted probabilities and interpretability.

The report is organised as follows. First, ordinal data and self-reported health are described. Next, the NHANES dataset and selected variables are introduced. Exploratory data analysis is then used to study the outcome and predictors. The methodology section explains cumulative logit ordinal regression, the parallel model and the non-parallel model. The results section presents fitted model outputs, odds ratios, p-values, predicted probabilities and an example prediction for a particular person. Finally, the report discusses the model choice, limitations and possible future work.

Ordinal Data

Ordinal data are categorical data with a meaningful order. In this project, the outcome variable is self-reported general health. The categories are:

Excellent, Very good, Good, Fair and Poor.

These categories clearly have a hierarchy. Excellent represents the best health category and Poor represents the worst health category. The category order is important, but the response is still categorical. This is the main reason why ordinary linear regression is not suitable for the outcome variable.

A simple way to understand ordinal data is to think of the categories as labels with order. Numbers such as 1 to 5 can be used to label the categories, but these numbers are not the same as regular measurements. If Excellent is coded as 1 and Poor is coded as 5, it does not mean that Poor is five times worse than Excellent. The numbers show ranking, not exact measurement.

Ordinal regression is more suitable because it keeps the order of the categories without treating the response as continuous. This allows the analysis to use the information that Excellent is better than Very good, Very good is better than Good, and so on. Cumulative logit ordinal regression is a widely used method for ordered categorical outcomes (McCullagh 1980; Agresti 2010).

In this project, ordinal regression is used to model the relationship between self-reported health and four predictors: age, BMI, gender and current smoking status. The model helps explain whether these predictors are associated with better or poorer self-reported health.

NHANES Dataset

NHANES is the National Health and Nutrition Examination Survey. It is a health survey of people in the United States conducted by the National Center for Health Statistics, which is part of the Centers for Disease Control and Prevention. NHANES collects information using questionnaire responses, physical examination measurements and health-related variables (Centers for Disease Control and Prevention, National Center for Health Statistics 2024a).

The analysis uses the NHANES data available through the R package `NHANES` (Pruim 2016). The data used in this project are from survey cycles 2009–2012. These data include the variables needed for the analysis: self-reported general health, age, BMI, gender and current smoking status.

The dependent variable is self-reported general health. In the dataset, this is recorded as ordered categories. The categories used in this analysis are Excellent, Very good, Good, Fair and Poor. The predictors used in the model are:

- **Age:** numerical predictor.
- **BMI:** numerical predictor representing body mass index.

- **Gender:** categorical predictor.
- **Smoking:** categorical Yes/No variable showing current smoking status.

The data preparation code is shown below. The code creates the ordered health outcome, converts the predictors into the correct formats, selects the variables used in the model, removes incomplete cases and drops unused factor levels.

```
##           Health           Age           Gender           BMI           Smoking
## Excellent: 213   Min.    :20.00   female:1236   Min.    :15.02   No :1581
## Very good: 849   1st Qu.:35.00   male  :1649   1st Qu.:24.03   Yes:1304
## Good      :1228   Median  :49.00                   Median :27.70
## Fair      : 496   Mean    :49.25                   Mean   :28.56
## Poor      :  99   3rd Qu.:62.00                   3rd Qu.:32.00
##                               Max.    :80.00                   Max.    :67.83
```

After selecting the required variables and keeping complete cases, the modelling dataset contained participants aged 20 to 80 years. This age range was observed in the selected modelling dataset; it was not added as a separate manual restriction in the analysis. The model therefore examines how age is associated with self-reported health across the available age range in the data.

Table 1 describes how the variables were used in the analysis.

Table 1: Variables used in the ordinal regression analysis.

Variable	Role	Type	Coding_or_description
Health	Outcome	Ordered categorical	Excellent, Very good, Good, Fair, Poor
Age	Predictor	Numeric	Age in years
BMI	Predictor	Numeric	Body mass index
Gender	Predictor	Categorical	Female or male
Smoking	Predictor	Categorical	Current smoking status: No or Yes

Table 1 shows that the outcome variable is ordered categorical, while the predictors include both numerical and categorical variables. This mix of variables is appropriate for cumulative logit ordinal regression.

Missing values were checked for the selected variables before fitting the models. This step was included as a data quality check, even though no major missing-data cleaning was required in the final modelling dataset. Table 2 presents the missing-value summary for the selected variables.

Table 2: Missing-value summary for selected modelling variables.

	Variable	Missing_Count	Missing_Percent
Health	Health	0	0
Age	Age	0	0
Gender	Gender	0	0
BMI	BMI	0	0
Smoking	Smoking	0	0

As shown in Table 2, the selected variables Health, Age, Gender, BMI and Smoking had zero missing values in the final modelling dataset. This confirms that the models were fitted using complete cases for all variables included in the analysis.

Table 3 summarises the final modelling dataset. It shows the number of complete cases, the observed age range, the mean age, the mean BMI and the number of outcome categories.

Table 3: Summary of the final modelling dataset.

Item	Value
Complete cases	2885
Age range	20 - 80
Mean age	49.25
Mean BMI	28.56
Outcome categories	5

Table 3 shows that the final dataset contains 2,885 complete cases. The average age is approximately 49.25 years and the average BMI is approximately 28.56 kg/m². These values provide a basic description of the sample before fitting the ordinal regression models.

Exploratory Data Analysis

Before fitting the ordinal regression models, exploratory data analysis was carried out to understand the structure of the data. This step helped examine the distribution of the outcome variable, the distribution of the selected predictors, and the visible patterns between predictors and self-reported health. The graphs were created using the `ggplot2` package in R (Wickham 2016).

Frequency tables for categorical variables

Table 4 shows the frequency of each self-reported health category. This table is important because ordinal regression models can be affected by very small categories.

Table 4: Frequency of self-reported health categories.

Health category	Count
Excellent	213
Very good	849
Good	1228
Fair	496
Poor	99

As shown in Table 4, most participants reported their health as Good or Very good, while fewer participants reported Poor health. This pattern is important because the Poor category has fewer observations than the middle categories.

Table 5 shows the gender distribution in the modelling dataset.

Table 5: Frequency of gender categories.

Gender	Count
female	1236
male	1649

Table 5 confirms that both female and male participants are represented in the modelling dataset. Gender is therefore included as a categorical predictor in the model.

Table 6 shows the distribution of current smoking status.

Table 6: Frequency of current smoking status.

Current smoking	Count
No	1581
Yes	1304

As shown in Table 6, there are more non-smokers than current smokers. This is useful to know because the smoking coefficient in the model compares current smokers with non-smokers while holding other predictors constant.

Graphical exploration

Figure 1 shows the distribution of self-reported health. This figure provides a visual version of Table 3.

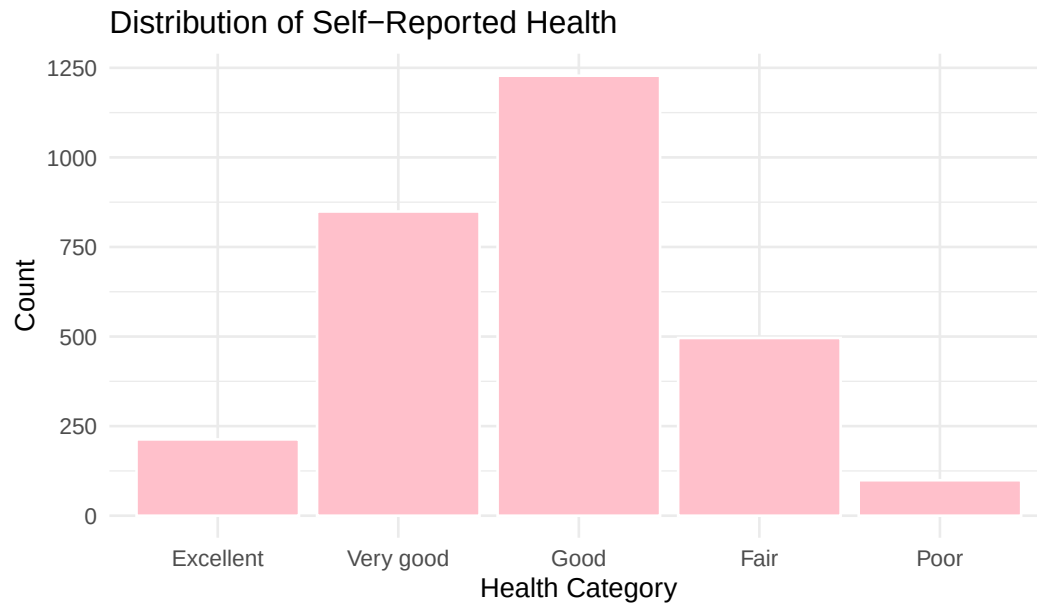


Figure 1: Distribution of self-reported health categories.

Figure 1 shows that most participants are in the Good and Very good categories. The smaller number of Poor cases is important because very small categories can make flexible models harder to estimate reliably.

Figure 2 shows the gender distribution in the modelling dataset.

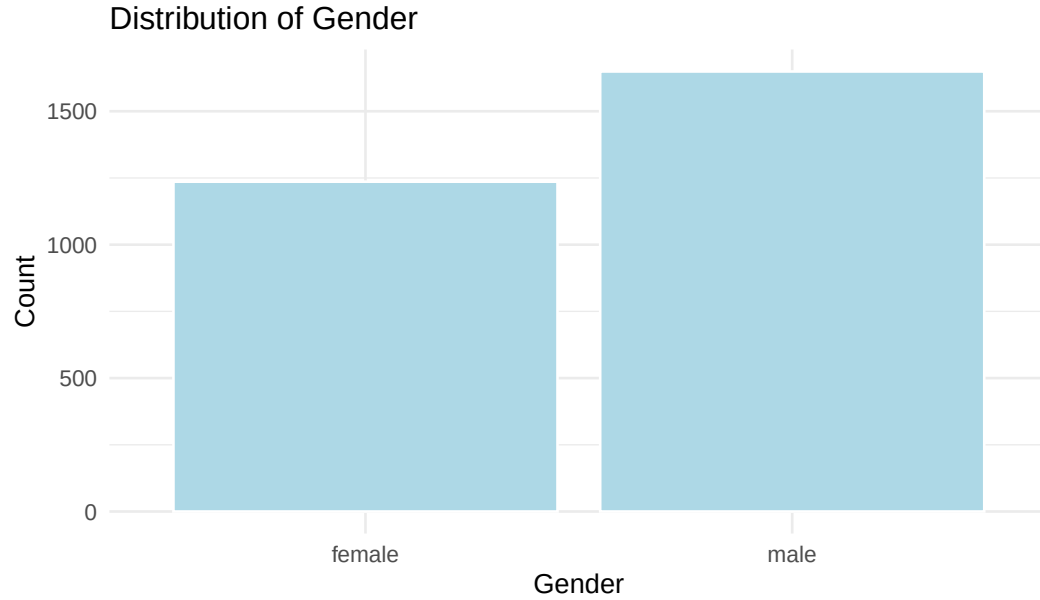


Figure 2: Distribution of gender in the modelling dataset.

Figure 2 shows that both gender categories are present in the data. This supports the use of gender as a categorical predictor in the ordinal regression model.

Figure 3 shows the distribution of current smoking status.

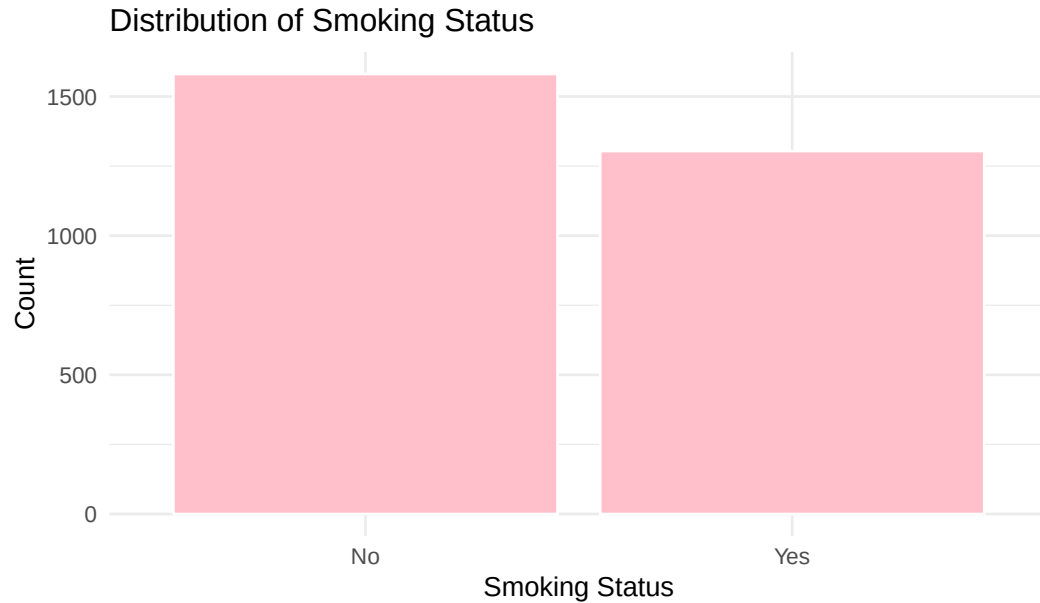


Figure 3: Distribution of smoking status.

Figure 3 shows that non-smokers are more common than current smokers. The model uses this variable to estimate whether current smoking is associated with poorer self-reported health after adjusting for age, BMI and gender.

Figure 4 shows the age distribution available in the selected modelling dataset.

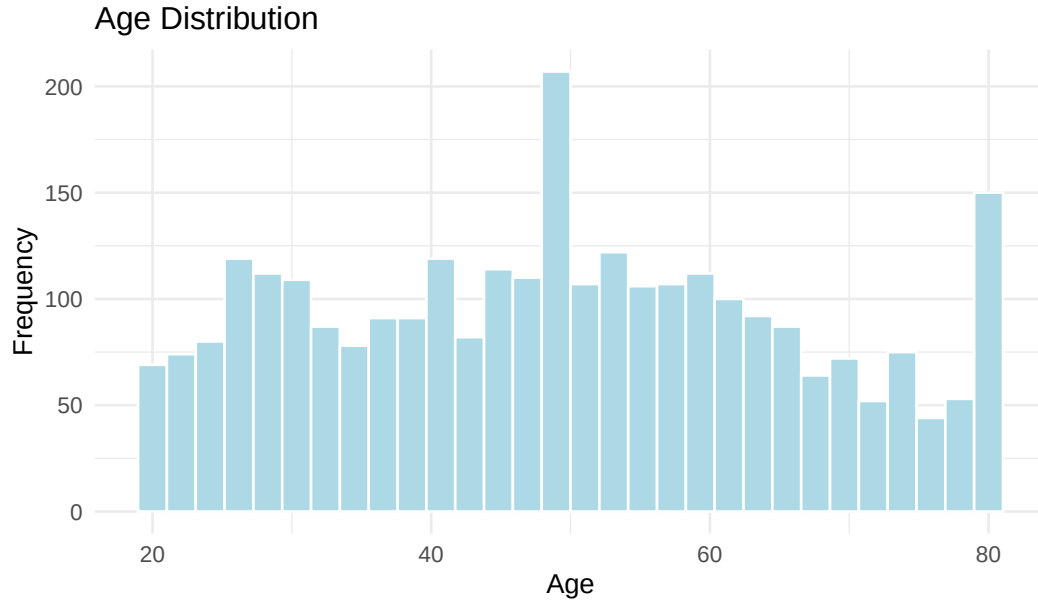


Figure 4: Age distribution of participants in the modelling dataset.

Figure 4 shows the observed age range in the modelling dataset. Age was included as a numerical predictor because one of the aims of the project was to understand how self-reported health is associated with age. Age was also used later in the predicted probability plots.

Figure 5 shows the BMI distribution.

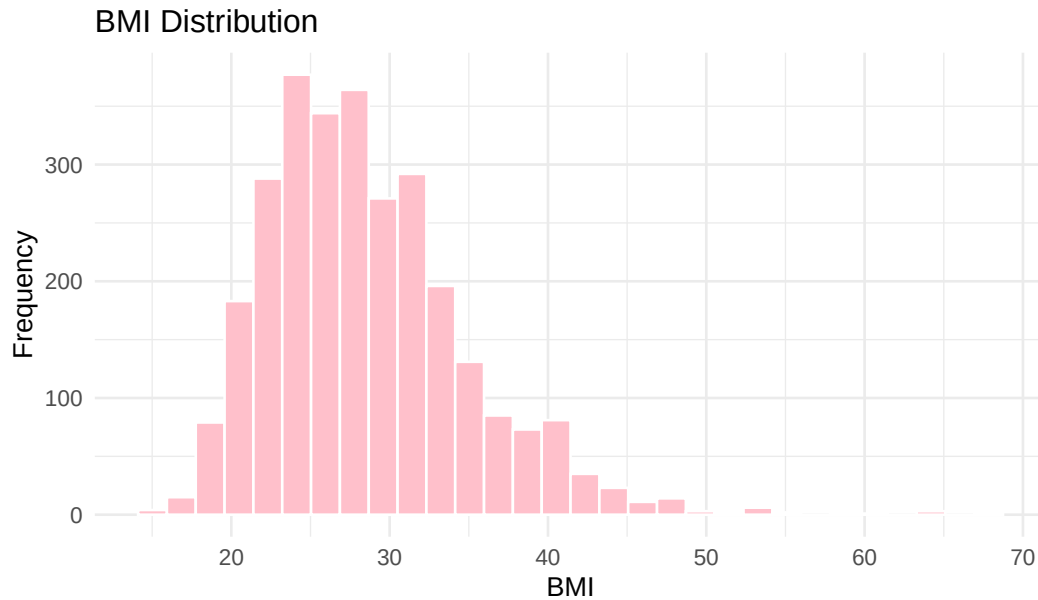


Figure 5: BMI distribution of participants in the modelling dataset.

Figure 5 shows that most BMI values are around the mid-to-high 20s, with some higher values. BMI was included as a continuous predictor because it is a common health-related measure and may be associated with self-reported health.

Figure 6 compares age across the five self-reported health categories.

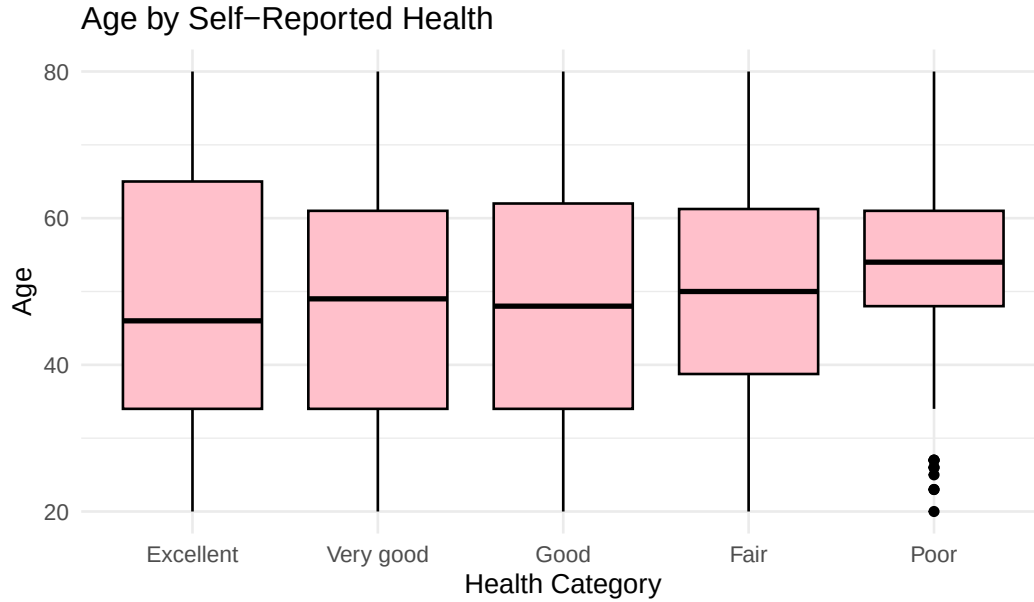


Figure 6: Age by self-reported health category.

Figure 6 suggests that older participants are more common in poorer health categories. This gives a visual reason for including age in the model.

Figure 7 compares BMI across the five self-reported health categories.



Figure 7: BMI by self-reported health category.

Figure 7 shows that BMI tends to be higher in the poorer self-reported health categories. This supports the inclusion of BMI as a predictor in the ordinal regression model.

Overall, the exploratory analysis shows that the outcome is ordered and that the selected predictors have visible relationships with self-reported health. This supports the decision to use cumulative logit ordinal regression.

Cumulative Logit Ordinal Regression

Ordered regression models are used when the response variable has categories with a natural ranking. The literature describes several approaches for ordered outcomes, including cumulative models, stage models and adjacent category models (Fullerton and Xu 2016; Course Material 2026a). In this project, the cumulative logit model was chosen because the aim was to model the ordered self-reported health outcome while keeping the interpretation clear and suitable for the project question.

Cumulative logit ordinal regression is used when the outcome variable has ordered categories. In this project, the outcome is self-reported health, ordered from Excellent to Poor. Since the response is categorical and ordered, cumulative logit ordinal regression is more suitable than ordinary linear regression.

The main idea of the model is to use cumulative probabilities. Instead of modelling only one health category at a time, the model considers the probability of being at or below a particular health threshold. For example, it can compare the probability of being in Excellent health versus all other categories, then Excellent or Very good versus the remaining categories, and so on.

For five ordered health categories, there are four cumulative thresholds. These thresholds split the ordered response into cumulative comparisons. In this project, the comparisons can be understood as moving across the health scale from the best category towards the poorest category. This is helpful because the model uses the full ordering of the response instead of ignoring the order or treating the categories as ordinary numbers.

For an ordered response variable Y , the cumulative logit model can be written as:

$$\log \left(\frac{P(Y \leq j)}{P(Y > j)} \right) = \alpha_j + \beta X$$

where Y is the ordered self-reported health outcome, j represents a health category threshold, α_j is the threshold-specific intercept, X represents the predictors and β represents the predictor effect.

In this project, the predictors are age, BMI, gender and current smoking status. The model estimates how these predictors are associated with the cumulative odds of reporting better or poorer self-reported health. The cumulative logit model is commonly used for ordinal response variables and can be fitted under a proportional odds, or parallel slopes, assumption (McCullagh 1980; Agresti 2010).

The coefficients from the model are estimated on the log-odds scale. Log-odds are useful for modelling, but they are not easy to explain directly. Therefore, the coefficients are converted into odds ratios by exponentiating them:

$$\text{Odds Ratio} = \exp(\beta)$$

An odds ratio helps explain the direction and strength of the relationship between a predictor and the outcome. If the odds ratio is greater than 1, the odds increase. If the odds ratio is less than 1, the odds decrease. In this project, because of the ordering and coding direction of the outcome, odds ratios below 1 indicate lower cumulative odds of being in better self-reported health categories.

For example, if the odds ratio for current smoking is below 1, it means that current smokers have lower odds of reporting better health compared with non-smokers, after adjusting for age, BMI and gender. This does not mean the probability directly decreases by the same percentage. It means the odds are lower on the odds scale.

The cumulative logit models in this project were fitted using the **VGAM** package and the `vglm()` function in R (Yee 2010, 2015, 2024). Two model forms were fitted: the parallel model and the non-parallel model. These two models are discussed in the next section.

Parallel and Non-Parallel Models

The cumulative logit ordinal regression model can be fitted in two main forms: the parallel model and the non-parallel model. Both models are designed for ordered categorical outcome variables, but they differ in the assumptions they make about the predictor effects.

In this project, the ordered outcome variable is self-reported health. The categories are Excellent, Very good, Good, Fair and Poor. The predictors are age, BMI, gender and current smoking status. Since the outcome

has a natural order, cumulative logit ordinal regression is appropriate. However, it is still necessary to decide whether the predictor effects should be assumed to be the same across all cumulative thresholds or whether they should be allowed to vary.

The parallel model assumes that each predictor has the same effect across all cumulative thresholds. This assumption is also known as the proportional odds assumption. Under this assumption, the model estimates one coefficient for each predictor. This means that age has one estimated effect, BMI has one estimated effect, gender has one estimated effect and smoking has one estimated effect across the ordered health outcome.

For example, the effect of BMI is assumed to be consistent across the cumulative thresholds. This makes the model easier to interpret because each predictor has only one overall odds ratio. The parallel model is therefore useful when the goal is to obtain a clear summary of the relationship between predictors and the ordered health outcome.

The parallel cumulative logit model was fitted in R using the `vglm()` function from the VGAM package. The model specification is shown below.

The argument `parallel = TRUE` tells R to fit the proportional odds version of the cumulative logit model. This means the same slope coefficient is used across the cumulative logits.

The non-parallel model relaxes the proportional odds assumption. Instead of forcing the effect of each predictor to be the same across all thresholds, the non-parallel model allows the effect of a predictor to differ from one threshold to another. This makes the model more flexible.

The non-parallel model was fitted using the same variables as the parallel model, but with `parallel = FALSE`.

The argument `parallel = FALSE` allows separate coefficients to be estimated for each cumulative threshold. This increases model flexibility, but it also increases model complexity. The output becomes longer because each predictor can have multiple threshold-specific coefficients.

Both models were included because they provide different information. The parallel model gives a simpler and more interpretable summary of predictor effects. The non-parallel model is useful as a sensitivity check because it examines whether allowing threshold-specific effects gives a noticeably better model.

A more flexible model is not always the best model. A model may have a slightly better fit but may also be harder to explain, less stable or less useful for interpretation. Therefore, the final model choice in this project considered both statistical fit and practical interpretation.

Fitted Model Results

This section presents the fitted model outputs. The focus is on coefficient estimates, odds ratios, p-values, residual deviance and predicted probabilities. The aim is not only to report the model output, but also to explain what the values mean in the context of self-reported health.

Parallel model output

The parallel model summary is shown using the code below. This output includes threshold estimates and predictor estimates.

```
##
## Call:
## vglm(formula = Health ~ Age + BMI + Gender + Smoking, family = cumulative(link = "logitlink",
##   parallel = TRUE), data = df, model = TRUE)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept):1  0.473418   0.211848   2.235   0.0254 *
## (Intercept):2  2.561695   0.210660  12.160 < 2e-16 ***
## (Intercept):3  4.590259   0.223261  20.560 < 2e-16 ***
## (Intercept):4  6.648358   0.247107  26.905 < 2e-16 ***
## Age           -0.011812   0.002196  -5.379  7.5e-08 ***
## BMI            -0.073276   0.005620 -13.038 < 2e-16 ***
```

```

## Gendermale      -0.153296   0.069597  -2.203   0.0276 *
## SmokingYes      -0.839791   0.075197 -11.168  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Names of linear predictors: logitlink(P[Y<=1]), logitlink(P[Y<=2]),
## logitlink(P[Y<=3]), logitlink(P[Y<=4])
##
## Residual deviance: 7418.731 on 11532 degrees of freedom
##
## Log-likelihood: -3709.366 on 11532 degrees of freedom
##
## Number of Fisher scoring iterations: 4
##
##
## Exponentiated coefficients:
##      Age      BMI Gendermale SmokingYes
## 0.9882572 0.9293442 0.8578759 0.4318007

```

The main predictor coefficients from the parallel model were extracted and converted into odds ratios. P-values were also included to identify which predictors showed statistical evidence of association with self-reported health.

Table 7: Parallel model coefficients, odds ratios, confidence intervals and p-values.

	Predictor	Estimate	Std_Error	z_value	Odds_Ratio	Lower_95_CI	Upper_95_CI	p_value_formatted
Age	Age	-0.0118	0.0022	-5.3789	0.9883	0.9840	0.9925	<0.001
BMI	BMI	-0.0733	0.0056	-	0.9293	0.9192	0.9396	<0.001
				13.0384				
Gendermale	Male	-0.1533	0.0696	-2.2026	0.8579	0.7485	0.9833	0.028
SmokingYes	Current smoker	-0.8398	0.0752	-	0.4318	0.3726	0.5004	<0.001
				11.1678				

Table 7 shows the main predictor results from the parallel cumulative logit model. The Estimate column gives the coefficient on the log-odds scale. The Odds Ratio column gives the exponentiated coefficient, which is easier to interpret. The p-value column shows whether there is statistical evidence that the predictor is associated with self-reported health after adjusting for the other predictors.

The age estimate is negative, meaning that increasing age is associated with lower cumulative odds of better self-reported health. The odds ratio for age is slightly below 1. This means that each one-year increase in age is associated with a small decrease in the odds of better self-reported health, holding BMI, gender and smoking status constant.

The BMI estimate is also negative. The odds ratio for BMI is below 1, meaning that higher BMI is associated with lower odds of better self-reported health, after adjusting for the other variables. This matches the exploratory analysis, where BMI tended to be higher among poorer health categories.

The male coefficient compares males with the reference gender category. The odds ratio below 1 suggests that males have lower cumulative odds of better self-reported health compared with the reference category, after adjusting for age, BMI and smoking.

The smoking coefficient is the strongest effect in the parallel model. The odds ratio for current smoking is clearly below 1. This means current smokers have lower odds of better self-reported health compared with non-smokers, after adjusting for age, BMI and gender. This is one of the most important findings of the model.

The hypothesis testing interpretation follows the usual rule: if the p-value is below 0.05, there is statistical evidence that the predictor is associated with self-reported health. In this model, age, BMI and smoking show statistical evidence of association, while gender is weaker compared with the other predictors. This matches the manual interpretation notes used during the project work (Project Calculation Notes 2026).

The fit statistics for the parallel model are shown in Table 7.

Table 8: Parallel model fit statistics.

Measure	Value
Residual deviance	7418.732
Log-likelihood	-3709.366
AIC	7434.732
BIC	7482.470

Table 8 shows the residual deviance, log-likelihood, AIC and BIC for the parallel model. These values are used to describe overall model fit. The AIC and BIC are useful because they consider both fit and model complexity. Lower values are generally preferred when comparing models fitted to the same data, although interpretation and stability must also be considered.

Non-parallel model output

The non-parallel model summary is shown below. This model gives separate threshold-specific coefficients.

```
##
## Call:
## vglm(formula = Health ~ Age + BMI + Gender + Smoking, family = cumulative(link = "logitlink",
##   parallel = FALSE), data = df, model = TRUE)
##
## Coefficients:
##      Estimate Std. Error z value Pr(>|z|)
## (Intercept):1  0.388767  0.412019  0.944 0.345391
## (Intercept):2  2.364726  0.246273  9.602 < 2e-16 ***
## (Intercept):3  4.279083  0.293258 14.592 < 2e-16 ***
## (Intercept):4  7.478113  0.482363 15.503 < 2e-16 ***
## Age:1         -0.006790  0.004191 -1.620 0.105196
## Age:2         -0.009686  0.002495 -3.882 0.000104 ***
## Age:3         -0.014748  0.002994 -4.926 8.40e-07 ***
## Age:4         -0.022341  0.005643 -3.959 7.52e-05 ***
## BMI:1         -0.075454  0.013147 -5.739 9.51e-09 ***
## BMI:2         -0.069742  0.006935 -10.057 < 2e-16 ***
## BMI:3         -0.060686  0.006863 -8.843 < 2e-16 ***
## BMI:4         -0.102471  0.008057 -12.718 < 2e-16 ***
## Gendermale:1  -0.110064  0.137152 -0.802 0.422266
## Gendermale:2  -0.128388  0.079680 -1.611 0.107118
## Gendermale:3  -0.195104  0.093889 -2.078 0.037707 *
## Gendermale:4   0.210609  0.182205  1.156 0.247725
## SmokingYes:1 -0.809197  0.152429 -5.309 1.10e-07 ***
## SmokingYes:2 -0.802619  0.086122 -9.320 < 2e-16 ***
## SmokingYes:3 -0.796958  0.100329 -7.943 1.97e-15 ***
## SmokingYes:4 -0.518359  0.196133 -2.643 0.008220 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Names of linear predictors: logitlink(P[Y<=1]), logitlink(P[Y<=2]),
## logitlink(P[Y<=3]), logitlink(P[Y<=4])
##
## Residual deviance: 7417.663 on 11520 degrees of freedom
##
## Log-likelihood: NA on 11520 degrees of freedom
##
## Number of Fisher scoring iterations: 2
##
```

```
##
## Exponentiated coefficients:
##      Age:1      Age:2      Age:3      Age:4      BMI:1      BMI:2
##      0.9932332    0.9903608    0.9853601    0.9779071    0.9273222    0.9326348
##      BMI:3      BMI:4 Gendermale:1 Gendermale:2 Gendermale:3 Gendermale:4
##      0.9411189    0.9026040    0.8957767    0.8795123    0.8227495    1.2344299
## SmokingYes:1 SmokingYes:2 SmokingYes:3 SmokingYes:4
##      0.4452156    0.4481537    0.4506980    0.5954972
```

The non-parallel coefficient table is longer because each predictor has coefficients for different cumulative thresholds.

Table 9: Non-parallel model coefficient output.

	Term	Estimate	Std_Error	z_value	Odds_Ratio	p_value_formatted
(Intercept):1	(Intercept):1	0.3888	0.4120	0.9436	1.4752	0.345
(Intercept):2	(Intercept):2	2.3647	0.2463	9.6021	10.6411	<0.001
(Intercept):3	(Intercept):3	4.2791	0.2933	14.5916	72.1742	<0.001
(Intercept):4	(Intercept):4	7.4781	0.4824	15.5031	1768.9005	<0.001
Age:1	Age:1	-0.0068	0.0042	-1.6202	0.9932	0.105
Age:2	Age:2	-0.0097	0.0025	-3.8819	0.9904	<0.001
Age:3	Age:3	-0.0147	0.0030	-4.9258	0.9854	<0.001
Age:4	Age:4	-0.0223	0.0056	-3.9593	0.9779	<0.001
BMI:1	BMI:1	-0.0755	0.0131	-5.7393	0.9273	<0.001
BMI:2	BMI:2	-0.0697	0.0069	-10.0572	0.9326	<0.001
BMI:3	BMI:3	-0.0607	0.0069	-8.8427	0.9411	<0.001
BMI:4	BMI:4	-0.1025	0.0081	-12.7178	0.9026	<0.001
Gendermale:1	Gendermale:1	-0.1101	0.1372	-0.8025	0.8958	0.422
Gendermale:2	Gendermale:2	-0.1284	0.0797	-1.6113	0.8795	0.107
Gendermale:3	Gendermale:3	-0.1951	0.0939	-2.0780	0.8227	0.038
Gendermale:4	Gendermale:4	0.2106	0.1822	1.1559	1.2344	0.248
SmokingYes:1	SmokingYes:1	-0.8092	0.1524	-5.3087	0.4452	<0.001
SmokingYes:2	SmokingYes:2	-0.8026	0.0861	-9.3196	0.4482	<0.001
SmokingYes:3	SmokingYes:3	-0.7970	0.1003	-7.9434	0.4507	<0.001
SmokingYes:4	SmokingYes:4	-0.5184	0.1961	-2.6429	0.5955	0.008

Table 9 shows the non-parallel model output. Unlike the parallel model, the non-parallel model provides threshold-specific effects. This means that age, BMI, gender and smoking can have different estimated effects at different cut-points of the outcome.

The advantage of the non-parallel model is flexibility. It can capture situations where a predictor has a stronger effect at one part of the health scale than another. For example, smoking may affect the probability of being in the best health categories differently from the probability of being in the poorest health category.

The disadvantage is that the output is harder to explain. Instead of one odds ratio for each predictor, there are several odds ratios for each predictor. This makes the model less simple and less suitable as the main interpretation model, unless the improvement in fit is clearly meaningful.

Model deviance comparison

Residual deviance was used to compare the parallel and non-parallel models. A lower residual deviance indicates better fit, but the size of the improvement must be considered. In likelihood-based modelling, deviance compares the fitted model with a saturated model, and values closer to zero indicate better fit relative to the best possible model (Course Material 2026b; Agresti 2015).

Table 10: Residual deviance comparison between the parallel and non-parallel models.

Model	Residual_Deviance
Parallel model	7418.732
Non-parallel model	7417.663
Difference	1.069

Table 10 shows that the non-parallel model has a slightly smaller residual deviance than the parallel model. The difference is only 1.068. Since both deviance values are above 7400, this is a very small improvement. This means that the non-parallel model does not provide a meaningful improvement in fit compared with the extra complexity it adds.

Understanding Estimates, Odds Ratios and P-values

This section explains the main model output in simple terms. It is included because model coefficients, odds ratios and p-values are often misunderstood.

Estimates

The Estimate value is the coefficient produced by the model. It is measured on the log-odds scale. A positive estimate means the predictor increases the cumulative log-odds. A negative estimate means the predictor decreases the cumulative log-odds.

In this project, the outcome categories are ordered from Excellent to Poor. Because of this coding direction, negative predictor estimates generally indicate lower cumulative odds of being in better health categories. For example, a negative smoking estimate suggests that current smokers are less likely to report better health compared with non-smokers, after adjusting for the other predictors.

The estimate is important because it gives the direction of the association. However, log-odds are difficult to interpret directly, so odds ratios are usually reported.

Odds ratios

An odds ratio is calculated by exponentiating the estimate:

$$\text{Odds Ratio} = \exp(\text{Estimate})$$

An odds ratio of 1 means no change in odds. An odds ratio greater than 1 means the odds increase. An odds ratio less than 1 means the odds decrease.

For example, if the smoking odds ratio is 0.432, this means current smokers have 0.432 times the cumulative odds of better self-reported health compared with non-smokers, after adjusting for age, BMI and gender. Another way to explain this is that current smokers have approximately $1 - 0.432 = 0.568$, or 56.8%, lower odds of better self-reported health. This is an odds interpretation, not a direct probability interpretation.

P-values

A p-value helps assess whether there is statistical evidence that a predictor is associated with the outcome. A small p-value means the observed association is unlikely to be due to random variation alone, assuming the model is correct. A common threshold is 0.05.

In this project, p-values below 0.05 are treated as evidence that the predictor is associated with self-reported health after adjusting for the other variables. However, a p-value does not measure the size of the effect. For effect size, the odds ratio is more useful.

A p-value also does not prove causation. This project uses observational survey data, so the results should be interpreted as associations rather than causal effects.

Example Prediction for a Particular Person

This section shows how the parallel cumulative logit model can be used for one person. The aim is to explain the practical meaning of the estimates, intercepts, exponentiation and predicted probabilities. This example follows the same manual calculation logic used in the project calculation notes (Project Calculation Notes 2026).

The example person is:

- Age = 27 years
- BMI = 10
- Gender = male
- Current smoking status = No

The BMI value of 10 is used only to reproduce the manual arithmetic example. It should not be treated as a typical adult BMI value. The purpose here is to show the calculation steps clearly.

Table 11: Illustrative person used for manual cumulative logit calculation.

Age	BMI	Gender	Smoking
27	10	male	No

Step 1: Extract model estimates

The parallel model has four threshold intercepts and one coefficient for each predictor. In the parallel model, the predictor coefficients are the same across the thresholds. Only the intercept changes from one threshold to another.

Table 12: Parallel model estimates used in the manual calculation.

Term	Estimate
(Intercept):1	0.473418
(Intercept):2	2.561695
(Intercept):3	4.590259
(Intercept):4	6.648358
Age	-0.011812
Gendermale	-0.153296
BMI	-0.073276
SmokingYes	-0.839791

Table 11 shows the estimates used in the calculation. The intercepts represent the four cumulative thresholds. The predictor estimates represent the contribution of age, gender, BMI and smoking status.

Step 2: Calculate the predictor part without intercept

The predictor part is calculated by multiplying each predictor value by its model estimate and then adding the results together. For this example person, male is coded as 1 and current smoking is coded as 0.

The calculation is:

$$(\beta_{Age} \times Age) + (\beta_{Male} \times Male) + (\beta_{BMI} \times BMI) + (\beta_{Smoking} \times Smoking)$$

Table 13: Manual calculation of the predictor part without intercept.

Component	Calculation	Value
Age part	-0.011812×27	-0.3189
Male part	-0.153296×1	-0.1533
BMI part	-0.073276×10	-0.7328
Smoking part	-0.839791×0	0.0000
Total predictor part	Sum of all parts	-1.2050

Table 12 shows the calculation of the predictor part. The total predictor part is the value before adding the threshold intercepts. This is similar to the console calculation where the age effect, gender effect, BMI effect and smoking effect are added first.

Step 3: Add each threshold intercept

The next step is to add each threshold intercept to the predictor part. This gives four cumulative logits, one for each threshold.

Table 14: Manual calculation after adding threshold intercepts.

Threshold	Intercept	Predictor_without_intercept	Cumulative_Logit
Threshold 1	0.4734	-1.205	-0.7316
Threshold 2	2.5617	-1.205	1.3567
Threshold 3	4.5903	-1.205	3.3853
Threshold 4	6.6484	-1.205	5.4434

Table 13 shows the four cumulative logits. The reason there are four logits is that the outcome has five ordered categories, so the model has four cumulative thresholds.

Step 4: Apply exponentiation

The cumulative logits are on the log-odds scale. To convert each logit into cumulative odds, the exponential function is applied:

$$\text{Cumulative Odds} = \exp(\text{Cumulative Logit})$$

Table 15: Conversion from cumulative logits to cumulative odds using $\exp()$.

Threshold	Intercept	Predictor_without_intercept	Cumulative_Logit	Cumulative_Odds
Threshold 1	0.4734	-1.205	-0.7316	0.4811
Threshold 2	2.5617	-1.205	1.3567	3.8834
Threshold 3	4.5903	-1.205	3.3853	29.5268
Threshold 4	6.6484	-1.205	5.4434	231.2270

Table 14 shows the result after applying $\exp()$. This step explains why exponentiation is used in ordinal regression. However, these values are cumulative odds, not final category probabilities.

Step 5: Convert cumulative logits to cumulative probabilities

The cumulative logit can also be converted into a cumulative probability using the logistic function:

$$P = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

where η is the cumulative logit.

Table 16: Conversion from cumulative odds to cumulative probabilities.

Threshold	Intercept	Predictor_without_intercept	Cumulative_Logit	Cumulative_Odds	Cumulative_Probability
Threshold 1	0.4734	-1.205	-0.7316	0.4811	0.3248
Threshold 2	2.5617	-1.205	1.3567	3.8834	0.7952
Threshold 3	4.5903	-1.205	3.3853	29.5268	0.9672
Threshold 4	6.6484	-1.205	5.4434	231.2270	0.9957

Table 15 shows the cumulative probabilities. These probabilities are cumulative, so they do not directly give the probability of one single health category. They describe the probability up to each threshold.

Step 6: Calculate final category probabilities

To decide which health category is most likely for this person, the probability of each individual category must be calculated. The easiest and safest way to do this in R is to use `predict()` with `type = "response"`.

Table 17: Predicted probability of each health category for the illustrative person.

Health_Category	Predicted_Probability
Excellent	0.3249
Very good	0.4704
Good	0.1720
Fair	0.0285
Poor	0.0043

Table 16 shows the predicted probability for each health category. The category with the largest predicted probability is the category the model considers most likely for this person.

Table 18: Most likely predicted health category for the illustrative person.

Health_Category	Predicted_Probability
Very good	0.4704

Table 17 shows the most likely predicted category. This final step is important because exponentiating the logits alone does not directly decide the health category. The correct approach is to calculate the probability of each category and then identify the category with the largest predicted probability.

This example shows the full logic of the model in a practical way. The process is:

1. multiply each predictor value by its estimate,
2. add the threshold intercepts,
3. convert logits into odds using `exp()`,
4. convert odds into cumulative probabilities,
5. calculate individual category probabilities,
6. select the category with the highest predicted probability.

This is why estimates, odds ratios and predicted probabilities all have different roles. Estimates are used inside the model, odds ratios explain predictor effects, and predicted probabilities help explain the expected health category for a specific person.

Non-parallel example for the same person

The same example person can also be checked using the non-parallel model. This is useful because the non-parallel model does not use one common effect for each predictor. Instead, age, BMI, gender and smoking can have different effects at each cumulative threshold.

The example person is kept the same:

- Age = 27 years
- BMI = 10
- Gender = male
- Current smoking status = No

The non-parallel calculation is different from the parallel calculation. In the parallel model, the predictor part was the same for all thresholds. In the non-parallel model, the predictor part is different for each threshold because each threshold has its own coefficient for age, BMI, gender and smoking.

For threshold j , the calculation is:

$$\eta_j = \alpha_j + \beta_{Age,j}(\text{Age}) + \beta_{Male,j}(\text{Male}) + \beta_{BMI,j}(\text{BMI}) + \beta_{Smoking,j}(\text{SmokingYes})$$

where η_j is the cumulative logit for threshold j . The subscript j shows that the coefficient can change from one threshold to another.

Table 19: Non-parallel model threshold-specific estimates used in the example calculation.

Term	Estimate
(Intercept):1	0.388767
(Intercept):2	2.364726
(Intercept):3	4.279083
(Intercept):4	7.478113
Age:1	-0.006790
Age:2	-0.009686
Age:3	-0.014748
Age:4	-0.022341
BMI:1	-0.075454
BMI:2	-0.069742
BMI:3	-0.060686
BMI:4	-0.102471
Gendermale:1	-0.110064
Gendermale:2	-0.128388
Gendermale:3	-0.195104
Gendermale:4	0.210609
SmokingYes:1	-0.809197
SmokingYes:2	-0.802619
SmokingYes:3	-0.796958
SmokingYes:4	-0.518359

Table 18 shows that the non-parallel model has more estimates than the parallel model. This is because each predictor can have a different estimate at each threshold. This makes the model more flexible, but also harder to explain.

The next table calculates the threshold-specific predictor part for the same person.

Table 20: Manual non-parallel calculation for the illustrative person.

Threshold	Age_part	BMI_part	Male_part	Smoking_part	Predictor_part	Intercept	Cumulative_Logit	Cumulative_Odds	Cumulative_Probability
Threshold 1	-	-	-	0	-1.0479	0.3888	-0.6592	0.5173	0.3409
Threshold 2	0.1833	0.7545	0.1101	0	-1.0873	2.3647	1.2774	3.5873	0.7820
Threshold 3	0.2615	0.6974	0.1284	0	-1.2002	4.2791	3.0789	21.7349	0.9560
Threshold 4	0.3982	0.6069	0.1951	0	-1.4173	7.4781	6.0608	428.7241	0.9977
	0.6032	1.0247							

Table 19 shows the manual non-parallel calculation. The important difference is that the predictor part changes across thresholds. This happens because the non-parallel model estimates separate effects for each threshold.

As with the parallel model, exponentiating the cumulative logits gives cumulative odds. The cumulative odds can then be converted to cumulative probabilities. However, these are still cumulative probabilities. To decide the most likely health category, the category probabilities must be calculated.

Table 21: Non-parallel predicted probability of each health category for the illustrative person.

Health_Category	Predicted_Probability
Excellent	0.3409
Very good	0.4411
Good	0.1740
Fair	0.0417
Poor	0.0023

Table 20 shows the predicted probability of each health category from the non-parallel model. The final predicted category is the category with the largest predicted probability.

Table 22: Most likely predicted health category from the non-parallel model.

Health_Category	Predicted_Probability
Very good	0.4411

Table 21 gives the most likely predicted category from the non-parallel model. This shows how the non-parallel model can also be used for individual-level prediction. However, because the non-parallel model uses different effects at different thresholds, the calculation is longer and more difficult to interpret.

This example supports the final model choice. The non-parallel model can produce predictions, but the explanation is more complicated. The parallel model is easier to explain because it uses one effect for each predictor. Since the non-parallel model did not meaningfully improve residual deviance, the simpler parallel model remains the preferred final model.

Predicted Probabilities

Predicted probabilities are useful because they show the model results in an easier form than log-odds or odds ratios. Instead of only saying that age or smoking changes the odds, predicted probabilities show how the probability of each health category changes across a predictor.

In this project, predicted probabilities were calculated across the observed age range. BMI was fixed at the mean value, gender was fixed at the reference category, and smoking was fixed as No. This makes the age pattern easier to view.

Figure 8 shows the predicted probabilities from the parallel model.

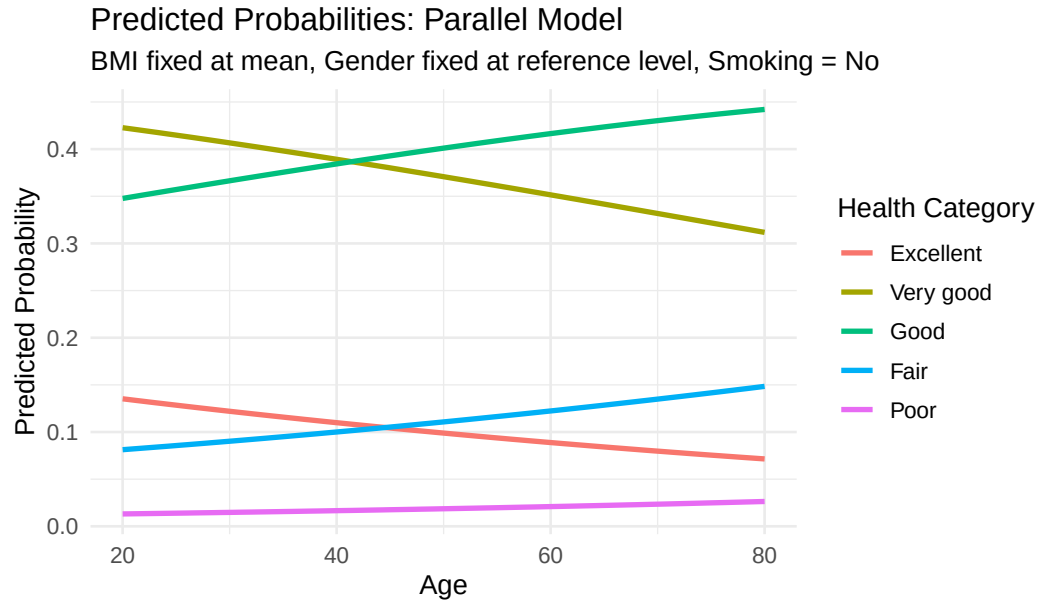


Figure 8: Predicted probabilities from the parallel model.

Figure 8 shows how the predicted probabilities change with age under the parallel model. The graph is easier to understand than log-odds because it shows probabilities for each health category directly.

Figure 9 shows the predicted probabilities from the non-parallel model.

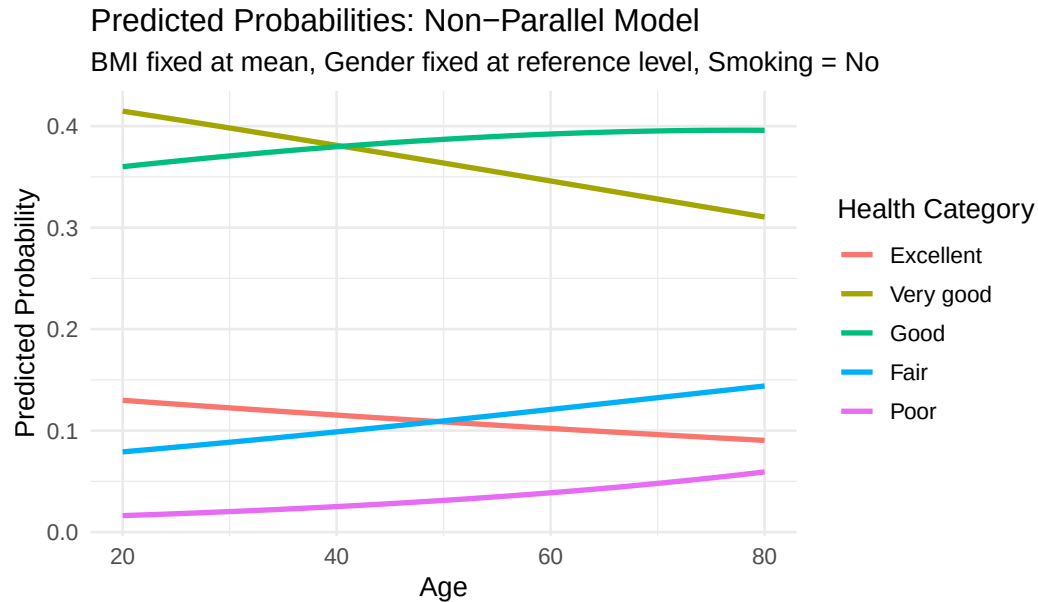


Figure 9: Predicted probabilities from the non-parallel model.

Figure 9 shows the predicted probability pattern under the non-parallel model. This model is more flexible, so the curves may differ slightly from the parallel model. However, flexibility alone is not enough to choose it as the final model.

Figure 10 compares the predicted probabilities from the parallel and non-parallel models side by side.

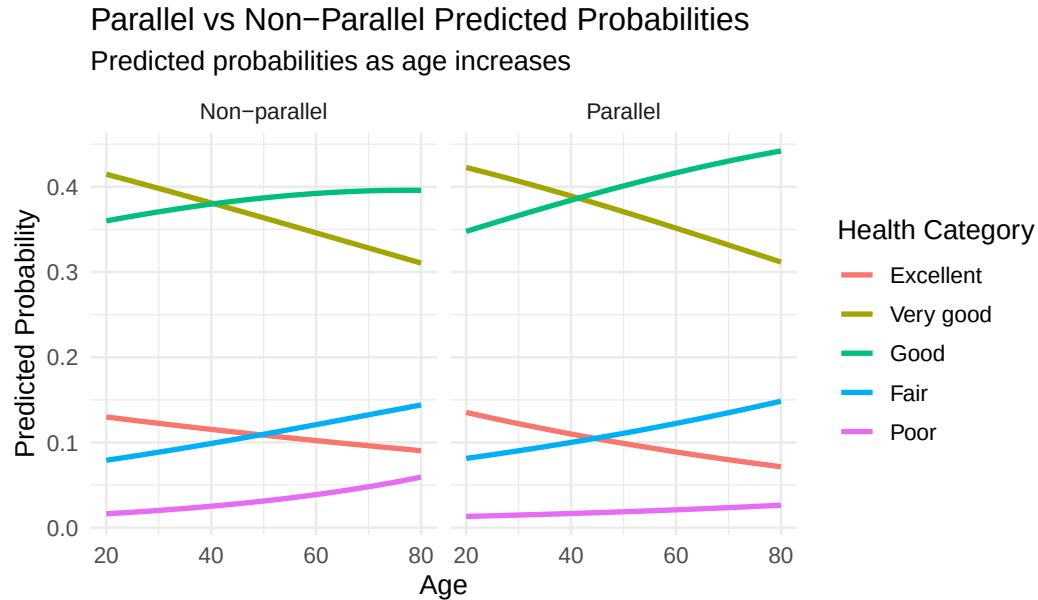


Figure 10: Parallel and non-parallel predicted probabilities compared.

Figure 10 shows that the predicted probability patterns from the two models are broadly similar. This supports the decision to keep the parallel model as the main model. The non-parallel model did not produce a large enough improvement to justify the extra complexity.

Model Comparison and Final Model Choice

The final model choice was based on both statistical results and interpretation. The non-parallel model had a slightly lower residual deviance than the parallel model. However, the reduction was only 1.068. Since the residual deviance values were above 7400, this difference was very small. This follows a common modelling principle: a more complex model should only be preferred when it provides a meaningful improvement in fit or interpretation (Course Material 2026b; Burnham and Anderson 2002).

The parallel model has several advantages. First, it gives one coefficient and one odds ratio for each predictor. This makes it easier to interpret the effect of age, BMI, gender and smoking. Second, it produced stable fit statistics such as residual deviance, log-likelihood, AIC and BIC. Third, the predicted probability patterns were similar to the non-parallel model.

The non-parallel model was useful as a sensitivity check. It showed that allowing threshold-specific effects did not meaningfully improve the model. However, the output was more complex because each predictor had separate coefficients at different thresholds. This made the model harder to explain clearly.

The final model decision is summarised by the code below.

```
##
## FINAL MODEL DECISION
## -----
## Parallel model residual deviance: 7418.732
## Non-parallel model residual deviance: 7417.663
```

Difference: 1.069

The non-parallel model has a slightly lower residual deviance.

However, the difference is very small compared with a deviance above 7400.

Therefore, the parallel cumulative logit ordinal regression model is preferred.

Based on the model comparison, the parallel cumulative logit model was selected as the main model for this project. It provided a good balance between statistical fit, stability and interpretability.

Discussion

The results show that ordinal regression is a suitable method for analysing self-reported health because the outcome is ordered. The analysis also shows that the selected predictors have meaningful relationships with self-reported health.

Age and BMI both had negative coefficients in the parallel model. This suggests that older age and higher BMI are associated with lower odds of better self-reported health. This is consistent with the exploratory graphs, where poorer health categories tended to include older participants and higher BMI values.

Current smoking had the strongest association among the predictors. The odds ratio for current smokers was below 1, suggesting that current smokers had lower odds of reporting better health compared with non-smokers, after adjusting for age, BMI and gender. This result is important because smoking is a major lifestyle factor related to health.

Gender was also included as a predictor. The coefficient for male participants compared with the reference category was negative in the parallel model. This suggests a difference in self-reported health by gender after adjustment for the other variables.

The comparison between the parallel and non-parallel models was an important part of the analysis. The non-parallel model was more flexible, but it did not substantially improve model fit. This supports the use of the simpler parallel model. In applied data analysis, the best model is not always the most complex model. A model should also be understandable, stable and useful for answering the research question.

A further strength of the report is that it includes both model-level interpretation and person-level prediction. The model-level interpretation explains the general association of age, BMI, gender and smoking with self-reported health. The person-level prediction example shows how the fitted model can be used for a specific profile. This makes the analysis easier to understand because it connects the statistical output to a practical calculation.

Limitations

This project has several limitations.

First, the analysis uses observational survey data. This means the results should be interpreted as associations, not causal effects. For example, the model can show that smoking is associated with poorer self-reported health, but it cannot prove causation from this analysis alone.

Second, the outcome variable is self-reported. Self-reported health is useful, but it is subjective. Different people may interpret the health categories differently. One person may describe their health as Good while another person with similar health conditions may choose Fair.

Third, only a small number of predictors were included. Age, BMI, gender and smoking are important, but many other variables may also affect self-reported health. These include income, education, physical activity, chronic disease, mental health, access to healthcare and diet.

Fourth, the project used complete cases after selecting the variables. Although the final modelling dataset had no missing values for the selected variables, complete-case analysis may still exclude some records from the original NHANES data.

Fifth, the analysis did not apply the full complex survey design features of NHANES, such as sampling weights, strata and clusters. Including survey design features would make the analysis more representative of

the United States population and would follow the full survey-analysis framework recommended for NHANES data (Centers for Disease Control and Prevention, National Center for Health Statistics 2024a, 2024b).

Finally, the non-parallel model was more complex and harder to interpret. Although it was useful as a sensitivity check, its additional complexity was not justified by the very small reduction in residual deviance.

Conclusion and Future Work

This project analysed self-reported general health using cumulative logit ordinal regression. The outcome variable was ordered from Excellent to Poor, so ordinal regression was more appropriate than ordinary linear regression. The predictors included age, BMI, gender and current smoking status.

The exploratory analysis showed clear patterns in the data. Most participants reported Good or Very good health, while fewer participants reported Poor health. Age and BMI also showed visible differences across health categories.

The parallel cumulative logit model showed that age, BMI, gender and smoking were associated with self-reported health. Odds ratios below 1 indicated lower cumulative odds of better self-reported health. Current smoking showed the strongest association with poorer self-reported health.

The non-parallel model was fitted as a sensitivity check. It produced a slightly lower residual deviance than the parallel model, but the difference was only 1.068. This was too small to justify the extra complexity. Therefore, the parallel model was selected as the final model because it was simpler, more stable and easier to interpret.

Future work could extend this project in several ways. First, additional predictors such as income, education, physical activity, chronic illness and mental health could be included. Second, NHANES survey weights and design variables could be used to make the results more representative. Third, interaction effects could be tested, for example whether the association between BMI and self-reported health differs by gender or smoking status. Fourth, partial proportional odds models could be explored as a middle option between the parallel and non-parallel models. Finally, an interactive dashboard could be developed to allow users to explore predicted probabilities for different age, BMI, gender and smoking profiles.

Overall, this project shows that cumulative logit ordinal regression is a useful and interpretable method for analysing ordered health outcomes. It also shows that model choice should consider both statistical fit and practical interpretation.

References

- Agresti, Alan. 2010. *Analysis of Ordinal Categorical Data*. 2nd ed. Wiley.
- Agresti, Alan. 2015. *Foundations of Linear and Generalized Linear Models*. Wiley.
- Burnham, Kenneth P., and David R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. Springer.
- Centers for Disease Control and Prevention, National Center for Health Statistics. 2024a. *National Health and Nutrition Examination Survey*. <https://www.cdc.gov/nchs/nhanes/>.
- Centers for Disease Control and Prevention, National Center for Health Statistics. 2024b. *NHANES Questionnaires, Datasets, and Related Documentation*. <https://www.cdc.gov/nchs/nhanes/>.
- Course Material. 2026a. *Introduction to Ordered Regression Models*.
- Course Material. 2026b. *Likelihood and Deviance*.
- Fullerton, Andrew S., and Jun Xu. 2016. *Ordered Regression Models: Parallel, Partial, and Non-Parallel Alternatives*. Chapman; Hall/CRC.
- McCullagh, Peter. 1980. "Regression Models for Ordinal Data." *Journal of the Royal Statistical Society: Series B (Methodological)* 42 (2): 109–42.

Project Calculation Notes. 2026. *Parallel Cumulative Logit Model Interpretation Notes*.

Pruim, Randall. 2016. *NHANES: Data from the US National Health and Nutrition Examination Survey*. <https://CRAN.R-project.org/package=NHANES>.

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer. <https://ggplot2-book.org/>.

Yee, Thomas W. 2010. “The VGAM Package for Categorical Data Analysis.” *Journal of Statistical Software* 32 (10): 1–34. <https://doi.org/10.18637/jss.v032.i10>.

Yee, Thomas W. 2015. *Vector Generalized Linear and Additive Models: With an Implementation in r*. Springer.

Yee, Thomas W. 2024. *VGAM: Vector Generalized Linear and Additive Models*. <https://CRAN.R-project.org/package=VGAM>.